

Mathematical overview

Greedy is not always minimum. For whole-grid axis-aligned square tilings, a 6-by-5 rectangle needs at least five squares, and five fit; the familiar biggest-square rule uses six. The contrast 4-by-3 needs four and greedy achieves four. Area and pairwise geometric fit supply lower bounds, not the greedy rule itself.

Restricted tiles: a 5-by-5 or 7-by-5 rectangle cannot be tiled with only side-2 and side-3 squares, although its area is a sum of 4s and 9s. A 6-by-5 can. Boundary decompositions and disjoint projections give obstructions that area alone misses.

Compressed recipes: consecutive equal square sizes are replaced by their counts. A nonsquare recipe $q_1, \dots, q_k$ with $q_k \geq 2$ determines the long/short ratio by the finite continued fraction $q_1 + 1/(q_2 + 1/(\dots + 1/q_k))$. All whole-grid rectangles with that recipe are integer enlargements of its coprime one. Sizes are lost; ratio is retained. A square has recipe [1].

Children compare a familiar rule with unrestricted choices, explain why permitted sizes may fail, and reconstruct a missing scale. Grades 2-3 can fit and count; Grades 4-5 can defend minima and reconstruction. All limits here require axis-aligned whole-grid squares with no gaps or overlaps. They make no claim about arbitrary tilted or noninteger squares.

Entry routes and readiness

Shared packet: Grades 2-5. The concrete entry requires fitting whole-grid squares, counting pieces and understanding the base square-removal action. Multiplication/area to 35 supports lower bounds. P4 requires distinguishing a square's size from a group's count; formal continued fractions are adult context.

Optional K-1 entry is covering a prepared small board, comparing piece counts or trying P3's allowed pieces. Do not offer the compressed recipe if the adult would have to choose and execute every step.

Materials and preparation

Per pair: duplicated rectangle boards, square pieces of sides 1-5, pencils, ruler and spare graph paper. Suggested stock: ten side-1, six side-2, four side-3 and two each side-4/5 pieces; reuse between trials. Working unit spacing is 10 mm. Print at 100% and precut with adult scissors if loose pieces are used; colored-pencil tile boundaries are a low-preparation alternative.

Keep the greedy reference beside the competing tiling. During P3 remove all pieces except side-2/3; side-1 squares would change the question. P4's blank work grids are capacities: draw a subrectangle that fits the recipe. Estimated preparation 15-25 minutes with reusable pieces. Exact physical fit and the procedure have not been pretested. Keep spare graph paper for enlarged P4 reconstructions beyond the printed 8-by-4 spaces. Five sets serve eleven children at three fixed tables, one adult per table; the upper trio rotates roles.

Launch together

Let children freely handle square pieces. Demonstrate one legal largest-square removal at one end of a non-task rectangle, leaving a rectangular remainder, then let a child try a free tiling on spare paper. Before P4, enact the printed 5-by-2 visual: two size-2 squares and two size-1 squares become count recipe 2,2. A count is not a side length.

First visit and return visits

P1-2: a 25-40 minute first visit. P3: a restricted-pieces return. P4: a 20-35 minute recipe visit. These are estimates. Save both greedy and best free tilings; proofs and reconstruction can continue later.

Problem 1-2: Biggest squares versus fewest squares

Entry, prerequisites and timing

Grades 2-3 keep two physical tilings and compare counts. Grades 4-5 defend P2's minimum. Allow 25-40 minutes. Pairwise fit arguments can be shown with paper squares; equations are optional.

Mathematical facts and solutions

P1, 6-by-5: greedy uses one side-5 square and five side-1 squares, six pieces. Five fit: two side-3 squares in a 6-by-3 strip, with three side-2 squares in the remaining 6-by-2 strip. Using $(x,y,side)$ from one corner, tiles are $(0,0,3),(3,0,3),(0,3,2),(2,3,2),(4,3,2)$. Their interiors are disjoint and areas total 30.

P1, 4-by-3: greedy uses one side-3 square plus three side-1 squares, four pieces, and this is optimal. One or two positive square areas cannot sum to 12. The only three-piece area decomposition is $4+4+4$, requiring three side-2 squares. Every pair overlaps in vertical projection because $2+2>3$, so horizontal projections must be disjoint, requiring width $6>4$. Thus three cannot fit.

P2: five is the minimum for 6-by-5. Tile sides can only be 1-5. One square does not have area 30; no two square areas sum to 30. The only three-square area decomposition is $25+4+1$. Side-5 and side-2 squares cannot be disjoint because their side sum 7 exceeds both board dimensions 6 and 5. The only four-square decomposition is $16+9+4+1$, where side-4 and side-3 likewise have sum 7. These area lists can be pooled from a finite square-area card search.

For disjoint axis-aligned squares, at least one pair of horizontal/vertical projections has disjoint interiors; if both projections overlap in their interiors, the squares overlap. This proves the geometric exclusions. The five-piece witness gives the matching upper bound, so the optimum is exactly five.

Hold-back hints

Ask which reference square causes the greedy count to grow. For a bound, list possible square areas, then check whether the largest two proposed pieces could fit side by side in either direction. Do not give the five-piece witness before experimentation.

Depth and further questions

Find another board where greedy is optimal and another where it fails. A failed search alone does not establish a minimum. Keep the whole-grid restriction explicit; allowing arbitrary square sizes requires a different argument.

Facilitator notes

The same 6-by-5 witness supports P3. It is not a fourth investigation, and children may reuse their saved tiling.

Problem 3: Allowed square sizes and hidden obstructions

Entry, prerequisites and timing

Grades 2-3 attempt all three boards using only side-2/3 pieces. Grades 4-5 turn failure into boundary or packing explanations. Allow 20-35 minutes. Adult may write area counts while children arrange tiles.

Mathematical facts and solutions

5-by-5: impossible. Area $25=4u+9v$ forces four side-2 squares and one side-3 square: v must be 1 modulo 4 and cannot exceed 2. Each horizontal boundary of length 5 must contain a side-3 edge, because side-2 edges alone sum to an even length. But the single side-3 square cannot touch both opposite boundaries 5 units apart. Thus the area-permitted collection does not fit.

6-by-5: possible by two side-3 and three side-2 squares, arranged in the two strips from P1. Area is $18+12=30$. Rotations/reflections of that tiling are accepted.

7-by-5: impossible. Area $35=4u+9v$ forces two side-2 and three side-3 squares. Any two side-3 vertical projections overlap because $3+3>5$; their horizontal projections must therefore be disjoint. Three horizontal intervals of length 3 need width at least 9, exceeding 7. No placement can avoid this obstruction.

Area divisibility and side-length decompositions are necessary but need not be sufficient. The impossible boards were chosen to reveal two different reasons: boundary demand and packing, not merely a shortage of pieces.

Hold-back hints

For 5-by-5 ask how an odd-length outside edge can be covered with lengths 2 and 3. For 7-by-5 try sliding two side-3 pieces vertically; can they avoid overlap without moving apart horizontally?

Depth and further questions

A strip of height 5 admits at most $\text{floor}(\text{width}/3)$ side-3 squares, since all their vertical intervals overlap pairwise. This gives a reusable packing bound. It is not a complete classification of every 2/3-square tiling.

Facilitator notes

Supply ample allowed pieces. A missing physical piece is not an impossibility proof.

Problem 4: Rebuild a size-free recipe

Entry, prerequisites and timing

Grades 2-3 draw or assemble from groups once the count convention is understood. Grades 4-5 explain scaling and unique ratio. Allow 20-35 minutes or a full return. No fraction-expression fluency is required.

Mathematical facts and solutions

Recipe 1,2: smallest whole-grid rectangle 3-by-2, square sizes 2,1,1. Recipe 2,1,2: 8-by-3, sizes 3,3,2,1,1. Recipe 1,1,3: 7-by-4, sizes 4,3,1,1,1. All fit as subrectangles in the supplied 8-by-4 work grids. The grids themselves are not the rectangles to be fully tiled.

Yes, different-sized rectangles share a recipe: 6-by-4 has recipe 1,2; 16-by-6 has 2,1,2; 14-by-8 has 1,1,3. Multiplying every length by the same positive integer preserves all size-group counts. Turning alone does not give a different-sized rectangle. End removal keeps each Euclidean remainder rectangular; the free tilings in P1 have no end-removal constraint.

For geometric reversal, choose final square side 1. The final group q_k is a q_k -by-1 strip. If the reconstructed remainder has long side s and short side t , the preceding group q rebuilds a rectangle with sides $q*s+t$ and s : attach q squares of side s beside that remainder. Repeat backward. For 2,1,2 this gives 2-by-1, then 3-by-2, then 8-by-3.

For a nonsquare rectangle the final count is at least 2. With one size group, the integer long/short ratio is at least 2. With several groups, the preceding distinct size is a larger positive integer multiple of the last, so their ratio is at least 2. The canonical count list determines a unique reduced ratio; integer side lengths are then integer multiples of that coprime pair. A list ending in 1 can be an alternate continued-fraction notation, but is not the distinct-size removal record used here. The only square recipe is [1].

Hold-back hints

Separate cards for group counts from square-size labels. Ask what happens if every drawn square is doubled. For reversal, an adult can hold a last-unit-square strip while the child decides where the next equal-square group belongs.

Depth and further questions

Give a partner a new positive recipe ending with a count at least 2, and reconstruct its reduced rectangle. Ask what extra information would determine actual scale: the last square size suffices. A full list of actual square sizes also fixes scale.

Facilitator notes

Check the final drawing by the forward biggest-square rule. Reconstruction without that check may accidentally build an arbitrary square tiling with the requested counts.

Sources, checks and classroom record

Sources and adaptation

Current Week 32, Squares inside rectangles, student packets pp. 1-5 and adult guide were inspected. The base studies last-square gcd and identical tiles; this encore studies algorithm optimality, restrictions and compressed reconstruction.

Euclid VII.2 is cited by the base materials for common-divisor context. Its text was not newly read here and is not the source of the new minimum-tiling claims. The area/fit and reconstruction arguments above are independently derived.

Laura Givental, Maria Nemirovskaya and Ilya Zakharevich, Math Circle by the Bay, Preface printed pp. viii-x (local PDF pages 9-11): themes with deep context and multiple entry levels, manipulatives, returning to explanations, and no universal estimate for a theme's duration. These are reported principles. The encore tasks, pairing and readiness routes here are local proposals, not classroom results from that book.

Verification

The author recomputed the facts, solutions and extension arguments from the actual tasks and design kernels. Independent mathematical reviews are retained separately with the editable sources.

Mathematical, render and extracted-source rebuild results belong in the release-validation record. No physical fit or procedure pretest and no classroom pilot has occurred.

What to record after a visit

Record each investigation actually used, examples attempted, a saved construction or plan, and where each pair wants to resume. A theme can support several visits; count use by investigation, not by finishing pages.

Record the child's action or words first; write your interpretation separately. Note whether children retained the choices after the launch or needed an adult to operate the investigation. All readiness and timing estimates below are unpiloted hypotheses.

Observe whether greedy and competing tilings are preserved separately, whether physical piece shortage is mistaken for impossibility, and whether recipe counts are confused with side lengths.

Status

Draft bonus companion. Unpiloted. Digital and mathematical checks do not establish physical fit, procedural rehearsal, staffing success or classroom pacing. Physical pretests have not been performed. Base student pages and base facilitator guide are separate.